Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Cubes and Cuboids: Faces, Edges and Vertices

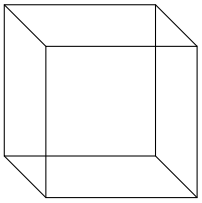
Three-dimensional shapes are shapes that have length, breadth and height. Because they have all three, they are called solid shapes. Some of them are regular, like a cube of sugar, a can drink or a match box; others are irregular, like a stone or a tree.

To describe any solid shape, you use three words:

- A face (or surface) is a flat or curved side of the shape.
- An edge is a line where two faces meet. It may be straight or curved.
- A vertex is a point corner where edges meet. The plural is vertices.

Count these parts on the cube in fig-01 and you will find its properties.

Visual



A cube drawn as a see-through solid: two equal squares, the nearer one offset down and to the right of the farther one, with the four pairs of corresponding corners joined.

Properties of a cube

A cube is a solid whose every face is a square. Look at the cube drawn in fig-01 — two equal squares joined by four straight edges.

1. It is a solid shape.
2. It has 6 square faces.
3. It has 12 edges.
4. It has 8 vertices.

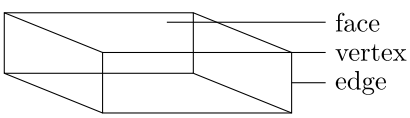
Properties of a cuboid

A cuboid is like a cube but its faces are rectangles, so it can be wide and shallow rather than equal in every direction. The cuboid in fig-02 shows you a face, a vertex and an edge.

1. It has 6 faces.
2. It has 12 edges.
3. It has 8 vertices.

So a cube and a cuboid both have 6 faces, 12 edges and 8 vertices. They differ in their faces: every face of a cube is a square, while the faces of a cuboid are rectangles.

Visual



A cuboid drawn the same way, wide and shallow, with three leader lines to the right naming the top face, the nearer top-right corner and the nearer right-hand vertical edge.

Let's check that you can find these properties on the picture. Take the cube in fig-01. Run your eye over it: count the squares that make it up — there are 6. Count the straight lines where two squares meet — there are 12 of

them. Count the point corners where three edges meet — there are 8. These three numbers, 6, 12 and 8, are the properties you should always be able to recall for both a cube and a cuboid.

Practice

1. How many vertices has a cuboid?
2. How many edges has a cube?
3. Copy the cuboid in fig-02 onto your paper and label one face, one edge and one vertex on it.

Section 2: Properties of Pyramids and Cones

A pyramid is a solid shape that narrows from a flat base up to a single point called the apex. A pyramid is named according to the shape of its base. For example, a pyramid whose base is a triangle is a triangular based pyramid, and one whose base is a square is a square based pyramid. Because the name always tells you the base, you can work out many of the properties without drawing anything.

Three things describe a solid shape:

- Faces — the flat surfaces that make up the shape.
- Edges — the lines where two faces meet.
- Vertices — the corners where edges meet (one corner is a vertex).

Here are the properties of four pyramids. Notice that the numbers depend on the base.

Shape	Faces	Edges	Vertices
Triangular based pyramid (tetrahedron)	4	6	4
Square based pyramid	5	8	5
Pentagonal based pyramid	6	10	6
Rectangular based pyramid	5	8	5

A triangular based pyramid has a triangle for its base, so it has four faces altogether: the triangular base plus three triangular sides. It is also called a tetrahedron.

Now consider the cone. A cone has a circular flat base and a curved face that rises to a point, the apex.

Its properties are:

- It has two faces — one curved face and one flat face (the circular base).
- It has one edge — the circular rim where the curved face meets the flat base.
- It has one vertex — the pointed apex.

So unlike a pyramid, a cone has only one flat face and only one vertex, because its side is curved rather than made of flat faces.

Worked example — naming and listing properties

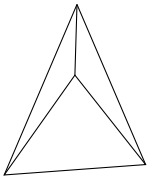
A solid has a pentagon for its base and a single apex above it. Name the solid and list its faces, edges and vertices.

Step 1. The name comes from the base. The base is a pentagon, so this is a pentagonal based pyramid.

Step 2. Look up its properties. From the table, a pentagonal based pyramid has 6 faces, 10 edges and 6 vertices.

Answer: Pentagonal based pyramid — 6 faces, 10 edges, 6 vertices.

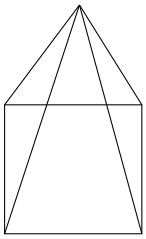
Visual



Triangular base Pyramid
(Tetrahedron)

A triangular base pyramid. The base triangle is drawn in perspective, its third vertex raised into the middle of the figure, and the apex above is joined to all three base vertices.

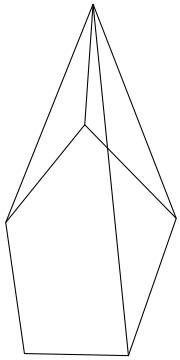
Visual



Square base Pyramid

A square base pyramid. The base is drawn as an upright rectangle and the apex above is joined to all four of its corners, so two edges run outside the rectangle and two cross it.

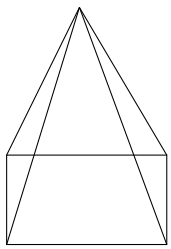
Visual



Pentagonal based
Pyramid

A pentagonal based pyramid. The base is drawn as a pentagon standing on one side with its fifth vertex uppermost, and the apex above is joined to that vertex, to the two vertices either side of it, and to the lower-right vertex — four edges from the apex, the last of them crossing the face of the solid.

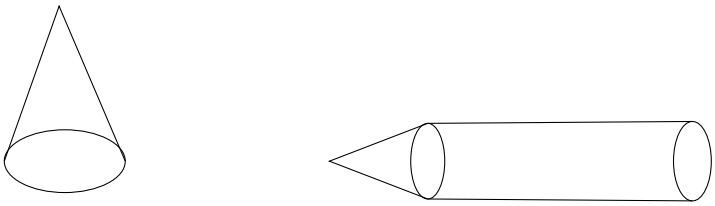
Visual



Rectangular based Pyramid

A rectangular based pyramid, drawn like the square one but on a base wider than it is tall.

Visual



Two drawings side by side. On the left an upright cone: an elliptical base with the two slant sides running up from the ends of the ellipse to the apex. On the right a cone lying on its side, apex at the left, its two slant sides reaching a narrow upright ellipse whose top and bottom then continue as two parallel lines to a second narrow ellipse further right.

Practice

1. A solid has a square for its base and a single apex above it. Name the solid and state how many faces, edges and vertices it has.
2. How many vertices does a tetrahedron have?
3. State the number of faces, edges and vertices of a cone. Identify which face is curved and which is flat.
4. A pyramid has 6 faces, 10 edges and 6 vertices. What is the shape of its base?

Check your work

Attempt every question before you read this page.

Cubes and Cuboids: Faces, Edges and Vertices

1. A cuboid has 8 vertices.

   A vertex is a point corner where edges meet. Count the corners of the cuboid: 4 on the nearer face and 4 on the farther face, giving 8 in all.
2. A cube has 12 edges.

   An edge is a line where two faces meet. On the cube, count 4 edges on the top face, 4 on the bottom face and 4 vertical edges joining them — 12 in all.
3. Your drawing should show a wide, shallow cuboid with a rectangle labelled 'face', one point corner labelled 'vertex' and one straight line where two faces meet labelled 'edge', exactly as in fig-02.

   Draw a cuboid the way fig-02 shows it: two rectangles, the nearer one offset down and to the right of the farther one, with the four pairs of corresponding corners joined. Then pick one flat side and label it 'face', pick one point corner and label it 'vertex', and pick one line where two faces meet and label it 'edge'.

Properties of Pyramids and Cones

1. Square based pyramid; it has 5 faces, 8 edges and 5 vertices.

The name comes from the base shape, so a square base gives a square based pyramid. From the properties table, a square based pyramid has 5 faces, 8 edges and 5 vertices.

2. 4 vertices.

   A tetrahedron is another name for a triangular based pyramid. Its properties are 4 faces, 6 edges and 4 vertices, so it has 4 vertices.

3. A cone has 2 faces, 1 edge and 1 vertex. It has one curved face and one flat face (the circular base).

   A cone has a flat circular base and a curved side that rises to a point. The curved side and the flat base make 2 faces. The circular rim where they meet is 1 edge, and the pointed apex is 1 vertex.

4. Its base is a pentagon, so it is a pentagonal based pyramid.

   The properties 6 faces, 10 edges and 6 vertices match the pentagonal based pyramid in the table, and a pentagonal based pyramid is named for its pentagon base.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-23ba9c8b	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____